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Actions of synthetic and endogenous steroids on the GABA_A receptor

J. J. Lambert, J. A. Peters and G. A. Cottrell

Traditionally steroid hormones have been thought to exert their effects on the mammalian CNS via an interaction with intracellular receptors, producing gene-regulated changes in protein synthesis. Whilst this is undoubtedly the mechanism of action for many steroids, some endogenous and synthetic steroids induce hypnosis/anaesthesia rapidly, suggesting an alternative mechanism of action. Jeremy Lambert, John Peters and Glen Cottrell review recent evidence which clearly demonstrates that some steroids are potent stereoselective potentiators of the actions of GABA at GABA_A receptors in vitro. Furthermore, at higher concentrations these steroids can directly activate the GABA_A receptor. As a number of the active steroids occur endogenously, the fascinating possibility exists that they may influence GABA_A receptors under physiological and pathophysiological conditions and so modulate the activity of the CNS.

It is now sixty years since Cashin and Moravsek¹ demonstrated the central depressant activity of intravenously administered cholesterol. In the intervening years it has become apparent that a hypnotic/anaesthetic property is common to many biogenic steroids². The pioneering investigations of Selye^{3,4} and subsequently Figdor *et al.*⁵ identified deoxycorticosterone acetate and some metabolites of progesterone possessing a saturated ring system, as particularly effective anaesthetics, exceeding the barbiturate thiopentone in their potency. Following these observations, attempts to formulate a steroidal anaesthetic suitable for surgical anaesthesia in man resulted in the development of alfaxalone (5 α -pregnane-3 α -ol,11,20,dione), the major active constituent of Althe-

sin used clinically, until recently, for the induction of anaesthesia⁶.

Biogenic and synthetic steroids modulate the GABA_A receptor

The anaesthetic effect of alfaxalone occurs rapidly, suggesting that it is not brought about by an interaction with intracellular receptors producing gene regulated changes in protein synthesis, a major site of action of steroid hormones⁷. The more plausible hypothesis of Scholfield⁸, that alfaxalone may induce anaesthesia by enhancing inhibitory transmission within the CNS, has recently received considerable support. Electrophysiological experiments utilizing the rat cuneate slice preparation have clearly demonstrated that alfaxalone potentiates extracellularly recorded responses to γ -amino-butyric acid and the selective GABA_A receptor agonist muscimol, but is without effect on responses to glycine⁹. Additionally, alfaxalone was reported to selectively enhance the binding of [³H] muscimol to synaptic membranes prepared from whole rat brain⁹.

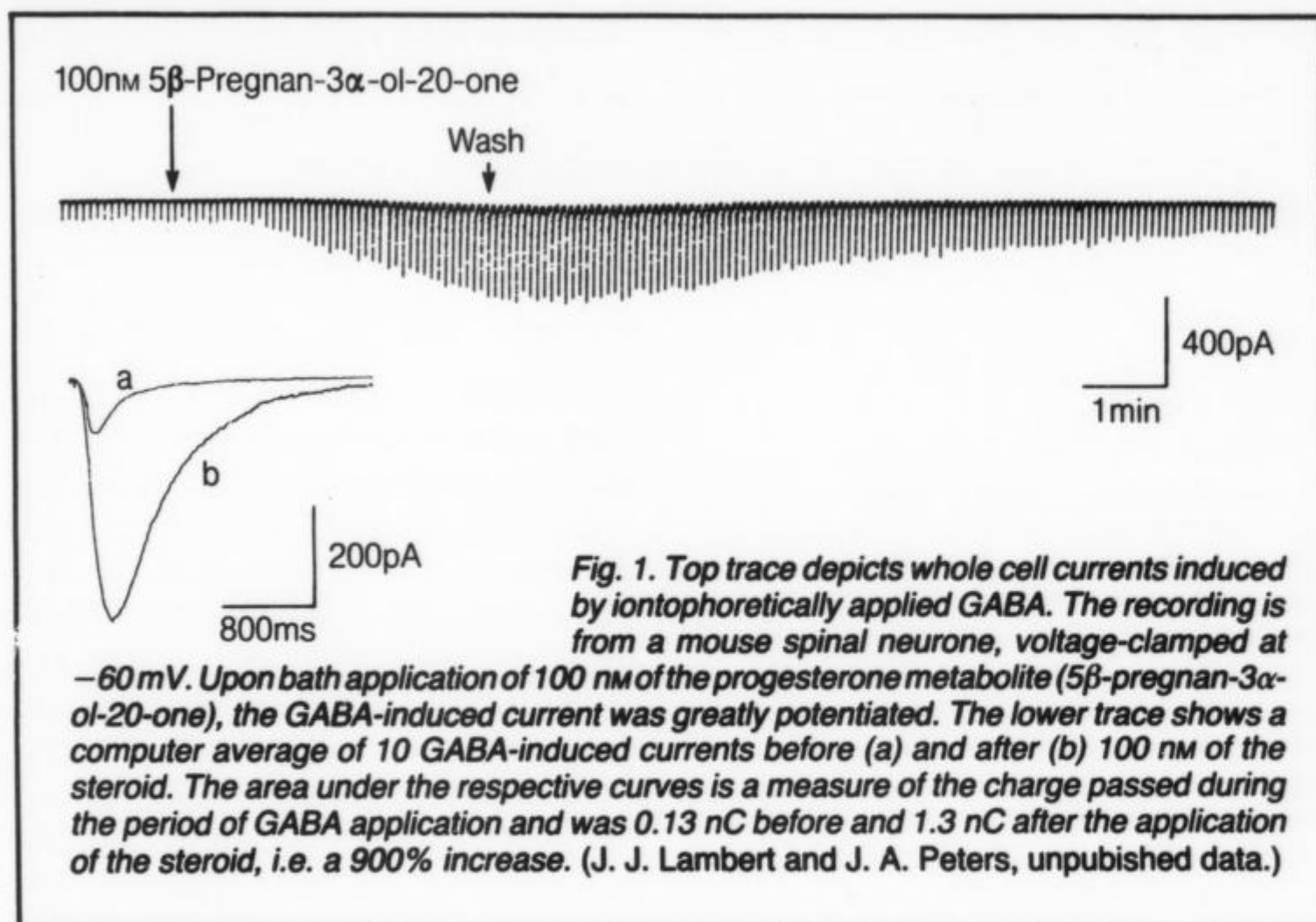
In corroboration of these results, recent experiments employing vol-

tage clamp techniques indicate that alfaxalone potentiates chloride currents evoked by GABA on rodent central neurones^{10,11} and adrenomedullary bovine chromaffin cells¹² maintained in cell culture. The steroid structure of alfaxalone is not incidental to such an action. The reduced metabolites of progesterone, 5 β -pregnan-3 α -ol-20-one and 5 α -pregnan-3 α -ol-20-one, have also been shown to be effective, at relatively low concentrations (30 nM-100 nM), in enhancing transmembrane currents elicited by GABA (Fig. 1)^{10,13,14}. Nor is enhancement of GABA-mediated responses confined to some progesterone metabolites, as androsterone¹⁵ and a deoxycorticosterone metabolite are also effective¹⁶.

Steroidal modulation of the GABA receptor complex is stereoselective

The 3 β -hydroxy isomer of alfaxalone, betaxalone, which lacks anaesthetic potency, has been reported to be ineffective in potentiating responses to GABA in the rat cuneate slice preparation⁹. This observation has recently been confirmed in experiments performed under voltage clamp conditions on bovine chromaffin cells¹². Indeed, in the latter system betaxalone, at relatively high doses (10-100 μ M), was observed to reversibly depress responses to GABA. The potentiation of GABA by progesterone metabolites is also stereoselective. Hence 5 β -pregnan-3 β -ol-20-one and 5 α -pregnan-3 β -ol-20-one fail to mimic the pronounced potentiation observed with the 3 α -ol-pregnane isomers when applied to chromaffin cells in the same concentration range (30-100 nM)¹⁴. Furthermore, whereas 5 β -pregnan-3 α -ol-20-one and 5 α -pregnan-3 α -ol-20-one stimulate the binding of [³H] muscimol to crude synaptosomal membranes derived from pig cortex, the 3 β -ol-pregnane isomers have little effect¹⁴. The stimulation of binding observed with the active isomers is associated with an increase

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in the apparent number of high affinity binding sites for [3 H]muscimol¹⁴, an effect concordant with that previously observed for alfaxalone⁹.

The molecular mechanism mediating the steroid-induced potentiation of GABA responses

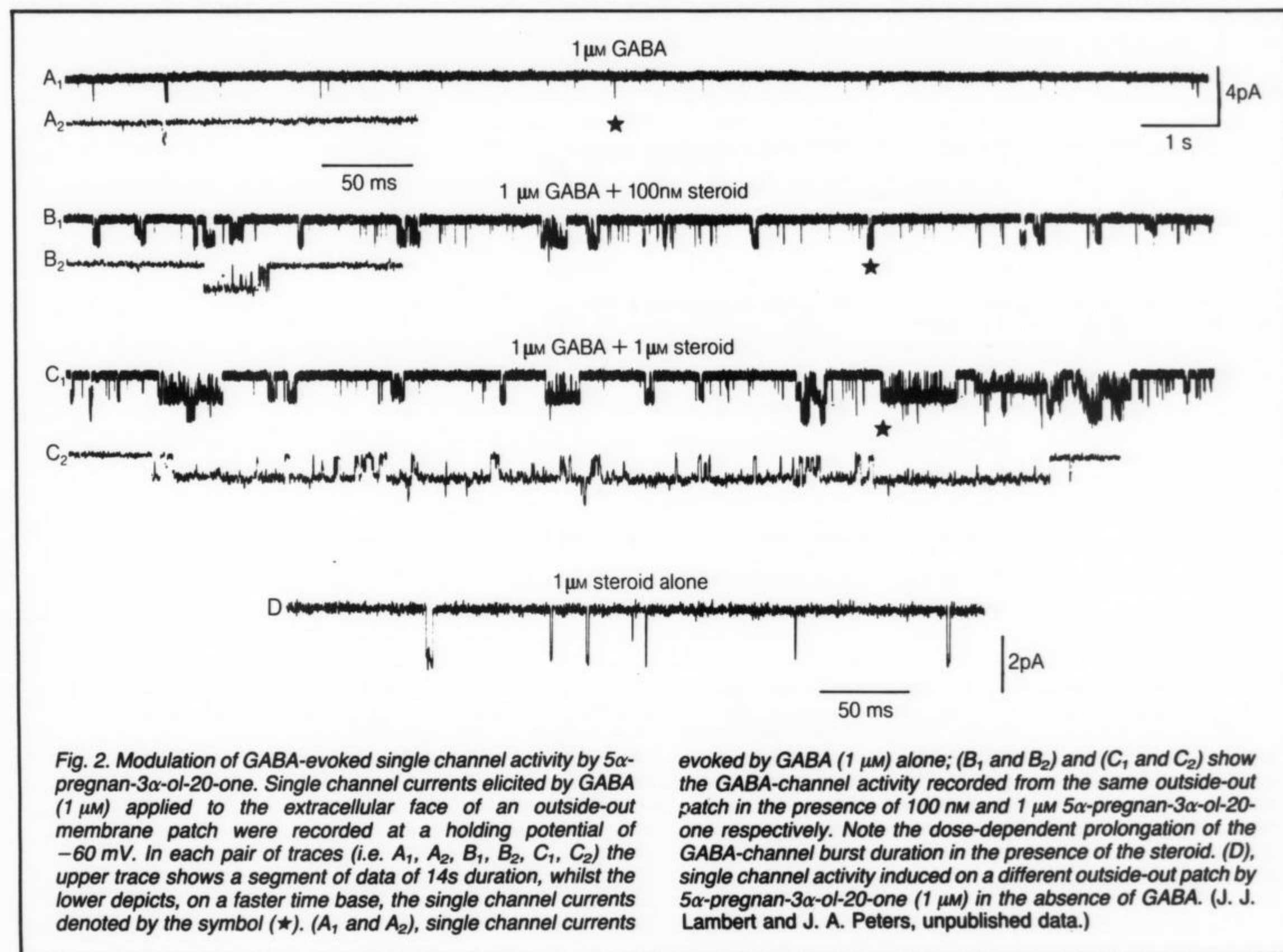
The mechanism by which some of the active steroids might act

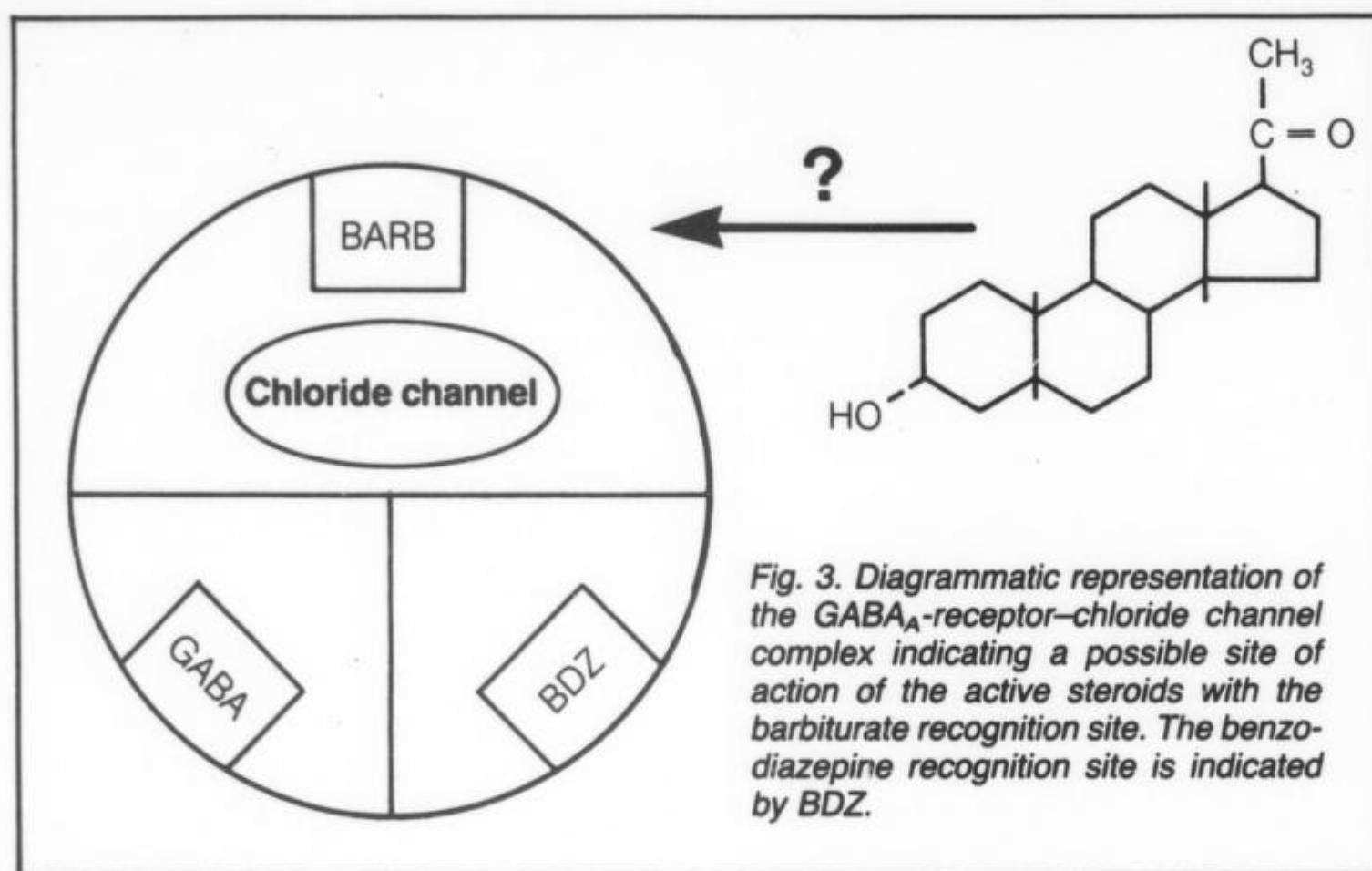
to enhance GABA-mediated responses has been investigated by current fluctuation and patch clamp techniques. The limited data presently available suggests that the steroids interact in some way with the GABA_A receptor-channel complex, to modulate the kinetic properties of single channel currents gated by GABA. Acting upon rodent spinal and

hippocampal neurones in culture, alfaxalone (1 μ M) significantly prolonged the mean open-time of GABA-gated ion channels, as inferred by fluctuation analysis¹⁰. The effects of 5 β -pregnan-3 α -ol-20-one upon GABA channel activity have been studied directly using outside-out membrane patches excised from bovine chromaffin cells. At concentrations previously demonstrated to be sufficient to elicit a substantial enhancement of the macroscopic current to GABA recorded from a whole cell, this steroid produced a marked prolongation of the burst duration of single GABA channel currents¹³. A similar effect of 5 α -pregnan-3 α -ol-20-one (100 nM–1 μ M) is illustrated in Fig. 2. Such a prolongation of GABA-channel burst duration underlies the potentiation of macroscopic GABA-evoked responses observed with these compounds.

Steroids directly activate the GABA_A receptor

At concentrations (1–100 μ M) greater than those required for potentiation of GABA, locally applied alfaxalone, a hydrocortico-





sterone metabolite, and some progesterone metabolites, but not betaxalone, also directly activate a transmembrane current^{12,13,16}. Such currents have a similar reversal potential to those activated by GABA and are reversibly antagonized by the GABA_A antagonist bicuculline^{12,13}. The active steroids also gate single ion channel currents of similar amplitude to those activated by GABA on outside-out membrane patches^{12,13} (see Fig. 2). The steroid-induced current can itself be potentiated by barbiturates, such as phenobarbitone¹¹⁻¹³ which prolongs the steroid single channel burst duration (Lambert & Peters, unpublished). Collectively these observations suggest that the active steroids can, in some way, directly activate the GABA_A receptor.

The site of steroid action

The remarkable potency of some steroids in potentiating the actions of GABA and their stereoselectivity suggests the existence of a specific site of interaction closely associated with the GABA_A receptor. It is established that there are specific binding sites for barbiturates and for benzodiazepines closely linked with the GABA_A receptor, and that some barbiturate and benzodiazepine drugs potentiate GABA responses evoked from central neurones (see Ref. 17) and bovine chromaffin cells¹⁸. An interaction between the steroids and the benzodiazepine site is unlikely, because the benzodiazepine antagonist Ro15-1788 (ethyl-8-fluoro-5,6-dihydro-5-methyl-6-oxo-4H-imidazo[1,5a][1,4]-benzodiazepine 3-carboxylate) does

not block the potentiating action of alfaxalone^{9,10,12} or 5 β -pregnan-3 α -ol-20-one on GABA responses¹³. Furthermore, in radioligand binding experiments, alfaxalone did not decrease, but actually increased diazepam binding⁹. Like the steroids, potentiation of GABA-evoked responses by some barbiturates is unaffected by low concentrations of Ro15-1788¹⁹. Further, both the barbiturate drugs and the active steroids displace the specific binding of the convulsant t-butylbicyclophosphorothionate (TBPS) to mammalian brain cell membranes¹⁶. Although some barbiturates and benzodiazepines potentiate GABA responses, fluctuation analysis suggests that each class of drug acts predominantly by a different mechanism. Pentobarbital for instance, like the active steroids, acts mainly by greatly prolonging the GABA channel open time, whereas some benzodiazepine drugs mainly increase the frequency of GABA-channel openings¹⁷. Thus the barbiturates and steroids may modulate GABA responses via a common mode of action or even via a common site of action, as represented diagrammatically in Fig. 3, although the latter seems less likely as steroid induced currents are greatly potentiated by barbiturates such as phenobarbital¹¹⁻¹³.

Thus the barbiturates and the steroids can, at least under some circumstances, exert their modulatory effects on the GABA_A receptor by actions at different sites. We are presently further investigating the interactions of steroids and barbiturates in an attempt to determine their site of action.

Pharmacological and physiological considerations

Are the actions of the steroids on the GABA_A receptor relevant to their anaesthetic action? A number of general anaesthetics with different structures to the steroids are known to potentiate GABA-mediated neurotransmission^{8,20} suggesting that such a mechanism may be involved in general anaesthesia. Further, 5 β -pregnan-3 β -ol-20-one, 5 α -pregnan-3 β -ol-20-one and betaxalone, which have little or no anaesthetic activity, did not potentiate GABA. The concentration of alfaxalone in the human brain in surgical anaesthesia is in the low micromolar range²¹. Hence, such concentrations are consistent with the proposal that an enhancement of GABA-mediated transmission may in part be responsible for the anaesthetic effects of the active steroids.

In-vivo administration of progesterone or deoxycorticosterone produces a number of metabolites, which can be detected in the brain (e.g. 2). Some of these metabolites are very active *in vitro* in potentiating GABA at GABA_A receptors. Hence these steroids could influence GABA_A receptors under physiological or pathophysiological conditions. Such actions could be especially important in conditions where steroid levels change dramatically e.g. in Cushing's Syndrome, or during pregnancy or stress, and may contribute to some of the behavioural changes known to occur in these conditions.

In summary, some steroids potently affect the activity of the GABA_A receptor-ion channel by potentiation, or in higher concentrations, by direct activation. It is suggested that some endogenous steroids may normally modulate GABA_A receptor-ion channel activity, under physiological conditions.

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Pharmacological and toxicological exploitation of amine transporters

Svante B. Ross

In these sophisticated days of advanced receptorology, a general view holds that factors such as uptake mechanisms have little interest for pharmacologists and toxicologists. But Svante Ross presents a new perspective. The re-uptake mechanisms in the membranes of the monoaminergic neurones, the amine transporters, are not specific for the transmitter amines but also transport compounds structurally related to the natural substrates. This makes the amine transporters suitable as specifically targeted delivery systems for drugs. On the other hand, such transporting systems also make these neurones vulnerable to toxic effects of compounds as a result of their neuronal accumulation.

The monoaminergic nerve terminals contain specific mechanisms for the re-uptake of the transmitter amines released by nerve stimulation. The functional role of these amine transporters is to terminate the action of the released amine molecules so that changes in the presynaptic activity can be detected by the postsynaptic receptors¹. These re-uptake mechanisms transport the bulk of the released amines back into the nerve terminals, where they are either restored in vesicles or metabolized by monoamine oxidase (MAO). The importance of the re-uptake mechanisms for the functional inactivation of the monoamines is shown by the fact that the most widely used antidepressant agents, e.g. imipramine and amitriptyline are potent inhibitors of

the re-uptake of noradrenaline and 5-hydroxytryptamine (5-HT).

Although the amine transporters are rather selective for the transmitter amines they are not specific for each transmitter amine. Thus, noradrenaline is transported by the dopamine transporter and vice versa. Some structural analogues of the biogenic amines can also be transported (for reviews see Refs 2-4). The lack of absolute specificities of the amine transporters makes these transport systems interesting as delivery systems for drugs designed to have selective actions within the aminergic neurones. The transporters may also take part in disease processes by accumulating toxic compounds into the neurones.

Characteristics of the amine transporters

The amine transporters are secondary active transport systems; i.e. the energy required for the transport of the amines against the concentration gradient is delivered from an ionic gradient over the membrane, in this case the

assymetric distribution of Na⁺ and K⁺ (Ref. 5). Studies of the amine transporters in isolated systems such as synaptosomes⁶, and membrane vesicles from platelets⁷ and pheochromocytoma cells⁸ have generated the scheme in Fig. 1 for the amine transport.

The first step in the translocation process is the binding of Na⁺ to the protein transporter. The protonated amine is then able to bind with high affinity to the transporter. Cl⁻ is also bound and is required for the net transport. This quaternary complex is then translocated in an as yet unknown manner and the amine, Na⁺ and Cl⁻ are dissociated at the inside, possibly due to a conformational change of the transporter protein induced by the high K⁺ concentration. The K⁺-transporter complex moves back to the outside, Na⁺ changes the conformation of the transporter so that it is able to bind the amine for another transport cycle. The translocation step through the membrane is generally believed to be the rate limiting one for the amine transport.

The amine transporters are systems with high affinities (apparent K_m 10⁻⁸ M-10⁻⁶ M) and low capacities. Thus uptake is saturated at low concentration of the amines. From the estimation of the number of the transporters (B_{max}) by radioligands [³H]imipramine and [³H]norzimeidine binding experiments and the maximal velocity of the transport (V_{max}) the molecular turnover numbers of the 5-HT transporter in porcine platelets⁹ and in synaptosomes from rat cerebral cortex¹⁰ have been estimated to be 300 and 60 min⁻¹, respectively. Thus, if all transporters take part in the translocation at saturating concentrations, the time for one translocation cycle in these preparations is 0.2 and 1 s, respectively.

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